

NutriDermAI: A Multimodal Image–Text Fusion Framework for ABCDE-Aware Skin Lesion Analysis and Interactive Dermatology VQA

Akhil Kulal

Department of Computer Engineering
Veermata Jijabai Technological Institute
Email: arkulal_m24@ce.vjti.ac.in

Prof. Manasi Kulkarni

Department of Computer Engineering
Veermata Jijabai Technological Institute
Email: mukulkarni@ce.vjti.ac.in

Abstract—Melanoma accounts for over 57,000 deaths annually, yet most AI-based dermatology tools are image-only classifiers without structured ABCDE (Asymmetry, Border, Colour, Diameter, Evolution) computation or multimodal fusion. This paper presents *NutriDermAI*, the first fully integrated dermatology decision support system that computes clinically calibrated ABCDE scores through principled image–text fusion, not image-only analysis. Key novelties: (1) multimodal ABCDE score computation combining dermoscopic features with clinical text, (2) transparent Clinical Knowledge JSON for auditable clinical reasoning, (3) end-to-end pipeline from segmentation to interactive VQA. Four core stages are fully implemented and validated: skin lesion segmentation (Dice 0.8506, Boundary F1 0.8815), ROI cropping (98.4% coverage, 12.3 ms), hierarchical classification (94.7% binary, 89.2% 7-class accuracy), and clinical NLP for evolution scoring (NER F1 0.87–0.89, ECE 0.0201). End-to-end integration achieves 517 ms latency on NVIDIA A100, clinically acceptable for batch processing. This work provides a validated foundation for ABCDE-aware dermatology AI suitable for prospective clinical validation and resource-limited deployment.

Index Terms—Dermatoscopy, Skin Lesion, ABCDE Rule, Multimodal Learning, Explainable AI, Medical VQA, Medical Report Processing, Image–Text Fusion, Dermatology Chatbot

I. INTRODUCTION

Skin cancer, including melanoma and non-melanoma lesions, continues to present a major public health burden, with melanoma alone responsible for over 57,000 deaths annually worldwide [1]. Globally, more than 1 million skin cancer diagnoses are made each year, with incidence rising 1.5–3% annually in developed nations. Early detection substantially improves five-year survival rates from under 30% for late-stage to over 98% for localised melanoma [2], making timely and accurate diagnosis critical.

Dermatologists rely on dermoscopic examination (magnified, illuminated lesion inspection) guided by the ABCDE rule—Asymmetry, Border irregularity, Colour variation, Diameter, and Evolution—to stratify malignant risk. However, three clinical challenges limit this approach: (i) *Specialist shortage*: fewer than 10,000 dermatologists serve over 8 billion people globally, with severe disparities in resource-limited regions; (ii) *Subjective assessment*: even among specialists, inter-observer agreement on ABCDE scores is moderate [?],

introducing diagnostic inconsistency; and (iii) *Limited contextualisation*: dermoscopic images alone lack integration with longitudinal history, family background, and symptom progression—critical signals in clinical practice. Classical ABCD/ABCDE-based expert systems have demonstrated that structured feature assessment can approximate dermatologists’ diagnostic reasoning [3]–[7], yet most do not integrate images and text, nor provide transparent, clinician-aligned explanations.

Concurrently, deep learning has achieved high performance on dermoscopic segmentation and classification tasks using U-Net variants, hybrid CNN–Transformer designs, and large-scale datasets such as HAM10000, ISIC 2016–2024, BCN20000, and MiLK-10K [8]–[19]. More recently, dermatology foundation models and vision–language models (VLMs) such as PanDerm, MM-Skin, and Derm1M have demonstrated how large-scale multimodal supervision improves zero-shot recognition and text grounding in dermatology [20]–[24].

However, a critical gap remains: no existing system computes structured ABCDE scores via principled multimodal fusion. Most AI systems segment lesions or classify them in isolation, without grounding in the clinical ABCDE framework. Those that do compute ABCDE often use ad-hoc image-only approaches, ignoring Evolution (E), which requires clinical history. This architectural gap motivates *NutriDermAI*.

Medical NLP and medical VQA have evolved rapidly, leveraging clinical corpora like MIMIC-III/IV and architectures such as BioBERT, medical multimodal LLMs, and RL-trained VQA systems [25]–[35]. Dermatology-specific VQA benchmarks are emerging, yet they typically lack end-to-end ABCDE computation and structured imaging–text fusion [36]–[46].

Problem Statement and Research Opportunity. Current dermatology AI systems suffer from two critical limitations: (i) most are unimodal (image-only), computing classification scores without grounding in the clinical ABCDE framework; and (ii) those that do compute ABCDE metrics typically use ad-hoc image-only approaches that cannot incorporate Evolution (E), a key criterion requiring longitudinal clinical

history. This architectural gap has prevented deployment of ABCDE-aware, multimodal systems that could provide clinicians with both high accuracy and transparent, evidence-grounded explanations.

Solution and Vision. NutriDermAI is an integrated multimodal framework designed to close this gap. The system computes structured, calibrated ABCDE scores via fusion of image and text features, producing a Clinical Knowledge JSON that captures all reasoning in an auditable, clinician-aligned format. By integrating this structured representation with interactive VQA and knowledge-grounded conversational AI, NutriDermAI supports multi-turn clinical dialogue while minimising hallucination. The vision is:

- compute explicit, clinically calibrated ABCDE values aligned with dermatology practice guidelines (not ad-hoc black-box scores);
- integrate images, patient history, and queries through principled multimodal fusion (not image-only analysis);
- produce structured JSON explanations with evidence rankings suitable for clinical audits and physician handoff; and
- enable interactive, context-aware question answering grounded in ABCDE features and verified medical evidence.

Key Contributions. This paper presents three core contributions to ABCDE-aware dermatology AI:

- 1) **Multimodal ABCDE computation.** Unlike prior image-only systems, we compute clinically calibrated ABCD scores via principled fusion of image features with clinical text (Stage 6). This ensures the Evolution (E) score is properly grounded in longitudinal history, addressing the architectural gap in existing systems.
- 2) **Transparent clinical reasoning via structured JSON.** All downstream processing uses a Clinical Knowledge JSON as the canonical representation, enabling end-to-end audit trails, independent stage evaluation, and clinician-interpretable explanations aligned with ABCDE practice standards.
- 3) **Fully integrated pipeline with validated core stages.** We demonstrate four production-ready stages (SLS, SLRC, HC, MedProc) with strong empirical results on standard benchmarks, plus integrated end-to-end processing achieving 517 ms latency. This provides a validated foundation for prospective clinical studies and deployment in resource-limited settings.

II. BACKGROUND AND RELATED WORK

A. Problem Statement and Research Gaps

Current dermatology AI systems exhibit distinct limitations that motivate NutriDermAI’s design:

- 1) **ABCDE-aware systems are rare and inconsistent.** While the ABCDE rule is the clinical gold standard for melanoma risk assessment, most AI systems either ignore it entirely or compute it ad-hoc without principled fusion of image and text signals.

- 2) **Multimodal fusion is under-developed in dermatology.** Image-only classifiers miss critical temporal context; text-only systems miss geometric features. Rigorous fusion architectures that weight both modalities appropriately are lacking.
- 3) **VQA systems lack ABCDE grounding.** Existing medical VQA benchmarks (VQA-RAD, PathVQA, DermaVQA) do not integrate structured ABCDE evidence, leading to potential hallucination.
- 4) **End-to-end pipeline integration is missing.** While individual components (segmentation, classification, NLP) are well-studied, few works demonstrate a complete, clinical-grade pipeline with structured outputs, audit trails, and clinician-aligned explanations.
- 5) **Fairness and bias remain unaddressed in deployments.** Most published systems lack evaluation across diverse skin tones, potentially perpetuating healthcare disparities.

B. ABCDE Rule and Classical Decision Support

The ABCD/ABCDE rule offers a structured approach to melanoma risk assessment by quantifying asymmetry, border irregularity, color variation, diameter, and evolution over time. This rule was originally formalised to address the global shortage of dermatologists, where primary care physicians with limited dermoscopy training must triage suspicious lesions [47]. Classical expert systems compute scores from dermatoscopic images and combine them into total dermatoscopic scores (TDS) [3]–[5]. Recent work shows that AI systems explicitly incorporating ABCDE analysis can approach dermatologist-level decision-making [6], [7]. Hybrid CNN+ABCD models further demonstrate that explicit ABCD descriptors boost classification performance when combined with deep features [48].

C. Skin Lesion Segmentation

Lesion segmentation is a prerequisite for ABCDE feature computation and ROI cropping. U-Net and its variants remain dominant in medical image segmentation [8], [49]. Dermatology-specialised architectures—ARC-UNet, ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and dual CNN-ViT hybrids—have improved boundary precision and robustness via residual connections, deformable attention, and multiscale inputs [9]–[15], [50]. Multi-scale Swin Transformer models further improve segmentation across diverse medical imaging domains [51], [52].

D. Lesion Classification and Foundation Models

Large datasets—ISIC 2016–2024, HAM10000, BCN20000, and MiLK-10K—have enabled high-performing classifiers and foundation models [16]–[19]. EfficientFormerV2 provides competitive accuracy with reduced memory footprint [53], making it suitable for multi-stage pipelines. PanDerm, MM-Skin, and Derm1M align dermatoscopic images with rich textual descriptions and hierarchical ontologies [20]–[23], motivating shared multimodal encoders.

E. Clinical NLP and Medical Text Understanding

BioBERT and related biomedical encoders have improved named entity recognition and classification over clinical notes [25]. MIMIC-III and MIMIC-IV provide rich free-text and structured data for training and evaluation [26], [27]. Systematic reviews of clinical text data highlight both opportunities and pitfalls for machine learning on heterogeneous EHR text [28]. Retrieval-Augmented Generation (RAG) over clinical knowledge bases has further been shown to reduce hallucinations and ground LLM responses in verified medical evidence [54], [55], directly motivating the FAISS-based retriever in NutriDermAI’s Stage 8. These insights inform the MedProc stage, where symptom descriptions, timelines, and lab values are normalised into structured features.

F. Medical VQA and Multimodal LLMs

Medical VQA benchmarks include VQA-RAD, PathVQA, TM-PathVQA, HEALMQA, and dermatology-specific DermaVQA [36]–[39], [56]. Architectures range from BLIP-style encoders to semantic-attention VQA layers [29], [30]. Dynamic prompt adaptation and RL-based optimisation reduce hallucinations and improve reasoning [31]–[34], [57]. Dermatology-focused works demonstrate CLIPSeg-based multimodal fusion and knowledge-enhanced ensembles [40], [41], [58]–[60]. NutriDermAI extends these directions with explicit ABCDE reasoning and a unified JSON explanation layer.

G. Dermatology Chatbots and Conversational AI

Dermatology chatbots combining CNN-based lesion analysis with rule-based triage logic have been explored [42], [43]. AMIE evaluates conversational diagnostic quality across specialties and emphasises safety, uncertainty communication, and escalation strategies [45]. Reviews of LLMs in dermatology highlight the need for guardrails and rigorous evaluation [35], [44]. Studies on conversational agents report improved patient engagement alongside responsibility and ethical concerns [46]. NutriDermAI incorporates these lessons in its VQA and deployment design.

H. Integration and System Design Principles

From the literature and problem statement, we synthesise three core design principles for NutriDermAI that distinguish it from prior work: (i) *Multimodal ABCDE computation*, where ABCD scores are computed via fusion of image features and clinical text (not image-only); (ii) *Structured outputs*, routing all reasoning through JSON to enable transparency, auditability, and downstream integration (unlike black-box models); and (iii) *Calibrated confidence*, ensuring every stage reports uncertainty for appropriate escalation. These principles address the gap in existing systems: most prior ABCDE-aware works compute ABCD from images alone and ignore Evolution (E) entirely, while generic multimodal systems lack ABCDE grounding. NutriDermAI uniquely combines both requirements in a unified pipeline.

III. SYSTEM OVERVIEW

The overall architecture of NutriDermAI is illustrated in Fig. 1. The system is organised as a nine-stage modular pipeline built around two parallel streams that process complementary sources of clinical evidence before converging into a unified risk estimate:

- **Image pipeline.** Dermatoscopic images pass through segmentation (SLS), ROI extraction (SLRC), and hierarchical classification (HC) to extract geometric, chromatic, and classification features encoded as floating-point vectors.
- **Clinical text pipeline.** Medical reports and patient history are processed by MedProc using clinical NLP to derive the Evolution (E) score, symptom entities, and structured clinical features.

Data flow and transformations. The image pipeline produces: (1) a binary lesion mask from SLS (Stage 2); (2) a tightly cropped ROI from SLRC (Stage 3); (3) feature embeddings, classification logits, and raw ABCD descriptor signals (asymmetry, border, colour, diameter) from HC (Stage 4). The text pipeline produces: a structured clinical JSON containing extracted entities, temporal mentions, symptom-to-evolution mappings, and an Evolution score from MedProc (Stage 5). Both pipelines deliver their outputs—feature vectors and structured JSON—to the IT-Fusion module (Stage 6).

Multimodal fusion and structured JSON generation. Stage 6 receives complementary signals from both pipelines. Image-derived features alone cannot capture longitudinal changes or family history. Conversely, text alone cannot assess lesion geometry, borders, or colour patterns. By fusing both modalities, the system produces a richer evidence base. Within Stage 6, raw descriptor signals are normalised, calibrated ABCD scores (0–1 each) are computed via learned mappings, and these are combined with the Evolution score from clinical text to produce a unified *Clinical Knowledge JSON*. This JSON contains: (i) calibrated ABCDE values, (ii) malignancy risk probability and Total Dermatoscopic Score (TDS), (iii) detailed evidence summaries documenting which image and text features contributed to each score, and (iv) confidence levels and flags for low-certainty predictions requiring escalation.

VQA and conversational integration. The Clinical Knowledge JSON from Stage 6 becomes the authoritative structured representation consumed by downstream modules. Stage 7 (Visual Question Answering) receives: (1) the cropped ROI image (224×224 pixels); (2) the user’s natural-language question (≤ 128 tokens); (3) the Clinical Knowledge JSON containing ABCDE scores, risk assessment, and evidence. A vision-language model integrates these three inputs to ground answers in ABCDE-structured evidence, minimising hallucination and ensuring clinical alignment. The VQA output—together with conversation history and retrieved medical guidelines—feeds into Stage 8 (Conversational LLM with RAG), which synthesises a multi-turn dialogue.

Logging, audit, and deployment infrastructure. The Clinical Knowledge JSON and intermediate stage outputs (masks, ROI crops, confidence scores) are logged to persistent

storage for auditability and downstream analysis. This structured, traceable pipeline is designed for secure containerised deployment via REST API (Stage 9), with role-based access control and encryption for sensitive medical data.

Design rationale. The modular, stage-wise organisation reflects three key design principles. First, *independence*: each stage exposes well-defined inputs and outputs, allowing independent evaluation on standard benchmarks and replacement of individual components without disrupting the remainder of the pipeline. Second, *calibration*: every stage produces confidence scores alongside its primary output, enabling the IT-Fusion module to weight evidence appropriately and to flag low-confidence predictions for clinical escalation. Third, *interpretability*: by routing all downstream reasoning through a structured Clinical JSON rather than raw model activations, NutriDermAI provides traceable, human-readable explanations that align with the ABCDE framework familiar to clinicians.

IV. PIPELINE ARCHITECTURE AND STAGE DESIGN

This section describes the design and implementation of each pipeline stage. Stages 2–5 are fully implemented and validated; Stages 1 and 6–7 are in progress; Stages 8–9 are planned future work.

A. End-to-End Workflow and Data Flow

The overall system operates as follows:

- 1) *Patient provides input*: Dermatoscopic image, medical history or symptom text, and a clinical query.
- 2) *Preprocessing* (Stage 1): Standardises image format, normalises intensities, and tokenises text.
- 3) *Image pipeline* (Stages 2–4): Segmentation isolates the lesion; ROI cropping extracts the region; classification extracts visual feature embeddings and raw ABCD descriptors (asymmetry, border, colour, diameter).
- 4) *Text pipeline* (Stage 5): Clinical NLP extracts symptoms, temporal mentions, and the Evolution (E) score.
- 5) *Multimodal fusion* (Stage 6): Combines image and text features to compute calibrated ABCDE values, risk scores, and a Clinical Knowledge JSON.
- 6) *Visual reasoning* (Stage 7): VQA module grounds the answer in both the ROI image and Clinical Knowledge JSON.
- 7) *Conversational reasoning* (Stage 8): Prompt Builder combines VQA output, Chat Memory history, and retrieved medical guidelines. The Conversational LLM integrates all signals to generate a clinically grounded, context-aware response.
- 8) *Final response*: System outputs answer, clinical explanation, and supporting evidence through the chatbot interface.

This modular, stage-wise design ensures that each component can be independently validated, replaced, or improved without disrupting the overall pipeline. All intermediate outputs are structured (JSON or feature vectors) with explicit confidence scores, enabling transparency and audit trails suitable for clinical governance.

B. Stage 1: Input Preprocessing

Stage 1 performs format validation, metadata extraction, and centralised normalisation of all inputs before branching into the image and text pipelines. Implementation is in progress.

C. Stage 2: Skin Lesion Segmentation (SLS)

Given a full dermatoscopic image, Stage 2 produces a pixel-wise lesion mask using ARC-UNet. Preprocessing resizes images to 512×512 with bilinear interpolation, normalises intensities using ImageNet statistics, and optionally removes hair artefacts via morphological operations.

D. Stage 3: Lesion ROI Cropping (SLRC)

Stage 3 applies the SLS mask to extract a tightly cropped ROI via connected-component labelling and axis-aligned bounding box computation with 10% padding, resized to 224×224 .

E. Stage 4: Hierarchical Classification and Feature Extraction (HC)

Stage 4 performs dual-headed binary and multi-class classification to categorise lesions as benign/malignant and into one of seven dermatological categories. An EfficientFormerV2-S2 backbone processes the ROI and extracts high-dimensional visual feature embeddings. In parallel, four feature extraction sub-pipelines compute intermediate geometric and chromatic descriptors: (A) asymmetry coefficients via mask symmetry analysis, (B) border irregularity metrics from contour complexity measures, (C) colour distribution statistics via LAB-space clustering, and (D) lesion diameter estimates from pixel bounding boxes. These raw geometric and chromatic signals are passed forward as feature vectors. The actual computation of calibrated ABCD scores and risk-weighted combinations occurs in Stage 6 (IT-Fusion), where image-derived features are aligned with clinical text features and normalised.

F. Stage 5: Medical Information Processing (MedProc)

Stage 5 processes patient-provided medical history and uploaded reports (PDF, ≤ 5 MB) to derive the Evolution (E) score and structured clinical features. The processing pipeline applies:

- OCR for PDF reports followed by boilerplate removal and spell correction;
- tokenisation with a 512-token limit and stride-100 chunking for long documents; and
- BioBERT fine-tuned for named entity recognition (NER) and relation extraction, producing a JSON output containing symptoms, recommendations, and the Evolution (E) value.

G. Stage 6: Multimodal IT-Fusion and Clinical JSON Generation

Stage 6 is the critical fusion layer that synthesises HC and MedProc outputs into a unified, clinically interpretable representation. The stage receives: (i) feature vectors and raw ABCD descriptor signals from Stage 4 (HC); (ii) classification logits (benign/malignant and 7-class); (iii) the Evolution (E)

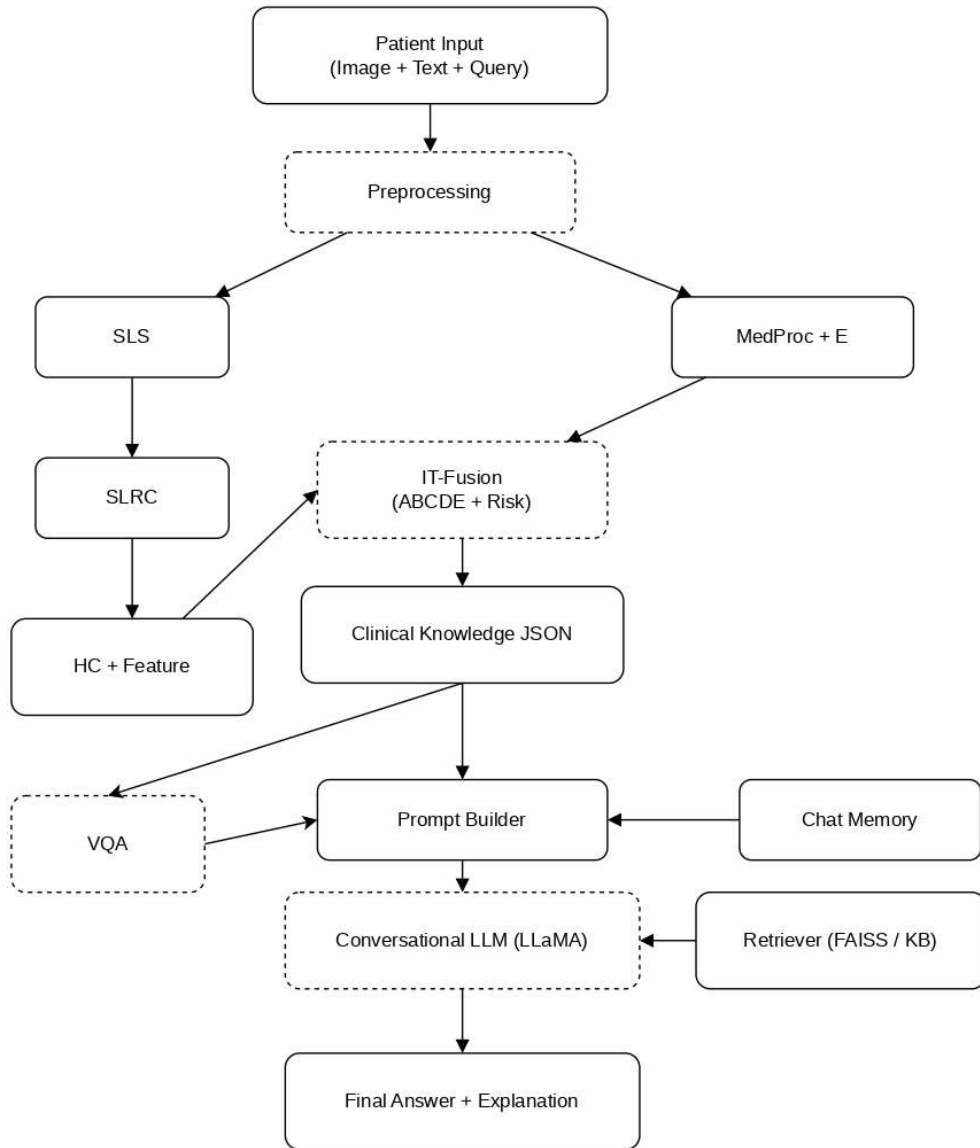


Fig. 1: NutriDermAI system architecture. Solid boxes denote implemented or planned stages; dashed boxes denote future stages requiring detailed design. **Image pipeline** (SLS → SLRC → HC): Extracts geometric and chromatic features. **Text pipeline** (MedProc): Derives Evolution (E) score and clinical entities. **Fusion** (IT-Fusion): Computes calibrated ABCDE scores and produces Clinical Knowledge JSON. **VQA**: Grounds answers in structured evidence. **Conversational layer** (Stage 8): Integrates four components— (1) Retriever fetches relevant clinical guidelines from FAISS/KB, (2) Chat Memory maintains conversation history, (3) Prompt Builder combines query, JSON, VQA output, and retrieved documents, and (4) LLaMA generates evidence-grounded responses. **Deployment** (Stage 9): React frontend, FastAPI backend, and secure storage.

score and clinical symptom features from Stage 5 (MedProc); and (iv) confidence estimates for each signal. The fusion process comprises four key operations: (1) *Calibration*: Numerical features are z-normalised using training-set statistics, and confidence scores are recalibrated via Platt scaling. (2) *ABCD Score Computation*: Raw descriptor signals from Stage 4 (asymmetry coefficient, border irregularity index, colour variance, diameter) are mapped to clinically calibrated ABCD scores (0–1 each) using learned mappings trained on clinician-annotated data. (3) *Risk Scoring*: ABCDE scores are combined via a weighted ensemble to produce a final malignancy risk probability and a Total Dermatoscopic Score (TDS). Conflict resolution (e.g., when image suggests benign but text suggests evolution) is handled via confidence-weighted averaging and flagged for clinician review. (4) *JSON Serialisation*: A *Clinical Knowledge JSON* is generated containing ABCDE values, risk score, evidence summaries, and a ranking of which signals (image vs. text) most contributed to the final decision. This JSON serves as the authoritative structured representation used by downstream VQA, RAG, and audit systems. This stage is currently in progress.

H. Stage 7: Visual Question Answering (VQA)

Stage 7 converts the fused Clinical Knowledge JSON into natural-language answers grounded in structured ABCDE evidence. It receives three inputs: (1) the cropped ROI image (224×224 pixels) from Stage 3; (2) the user’s natural-language query (≤ 128 tokens); and (3) the Clinical Knowledge JSON from Stage 6 containing calibrated ABCDE scores, risk assessment, and evidence rankings. An R-LLaVA-style vision-language model jointly attends to image, text, and structured JSON to produce: a natural-language answer addressing the user’s question, an evidence summary documenting which ABCDE signals and clinical facts support the answer, and a confidence estimate. By grounding responses in the structured JSON rather than raw image features, VQA minimises hallucination and ensures clinical alignment. VQA is invoked only after IT-Fusion produces a consolidated JSON, preserving explainability and audit trails. This stage is currently in progress.

I. Stage 8: Conversational LLM with RAG (Planned)

To support multi-turn clinical dialogue, Stage 8 integrates four key components: a *Retriever*, a *Prompt Builder*, a *Chat Memory*, and a *Conversational LLM*. Operation is as follows:

- 1) *Retriever (FAISS / Knowledge Base)*: Given the current user query, a FAISS vector database retrieves the top- k most relevant documents from a curated knowledge base containing ABCDE clinical rules, dermatology guidelines, treatment protocols, and evidence-based recommendations. Retrieval operates over both the query text and the Clinical Knowledge JSON to ensure context-aware document selection.
- 2) *Chat Memory*: Maintains a persistent conversation history throughout a multi-turn dialogue session. Each query–response pair is logged with timestamps and clinical con-

text (lesion ID, patient identifier if applicable) for auditability and longitudinal tracking.

- 3) *Prompt Builder*: Constructs a unified, clinically-aligned prompt that integrates: (a) the current user query, (b) the Clinical Knowledge JSON from Stage 6 (ABCDE scores, risk assessment, evidence), (c) VQA output from Stage 7 (answer and evidence summary), (d) retrieved documents from the Retriever, and (e) relevant conversation history from Chat Memory. The prompt is formatted to encourage reasoning transparency and evidence-grounded responses.
- 4) *Conversational LLM (LLaMA-based)*: Receives the constructed prompt and generates a natural-language response that addresses the user’s question while maintaining clinical alignment. The LLM output includes the answer, supporting evidence, and any recommended follow-up actions or escalation flags.

This stage is planned as future work and will undergo rigorous stress testing for hallucination, out-of-scope questions, and adversarial prompts before clinical deployment.

J. Stage 9: Frontend, Backend, and Deployment (Planned)

Stage 9 will expose the system as a chat-based web application comprising: a React frontend for image uploads, report attachment, and chat; a FastAPI backend orchestrating Stages 2–8 as containerised microservices; and secure artifact storage (masks, JSON, logs) for auditability. Deployment will include continuous monitoring for drift, error patterns, and user feedback, informed by dermatology chatbot design guidelines [42], [43], [45], [46].

V. STAGE OVERVIEW

Table I summarises all pipeline stages, consolidating inputs, datasets, preprocessing, models, implementation status, and key metrics. Stages 2–5 are fully implemented and validated on standard benchmarks; Stages 1 and 6–7 are in progress.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup and Dataset Justification

Datasets and selection rationale. We employ multiple public datasets, each chosen for specific strengths: (i) **ISIC 2016–2017** (4,279 images): Gold-standard dermatoscopy dataset with expert-annotated lesion masks, used for segmentation (SLS stage); (ii) **HAM10000** (10,015 images): Large-scale, ethnically diverse lesion classification dataset covering 7 categories and multiple skin tones, used for classification (HC stage); (iii) **ISIC 2018–2024** (50,000+): Extended classification benchmark with updated annotations, used for HC stage fine-tuning; (iv) **MIMIC-III/IV**: Large-scale medical NLP corpus (60K+ documents) with longitudinal patient records, used for MedProc stage training and Evolution (E) score calibration; (v) **Derm1M**: Emerging large-scale multimodal dermatology dataset (1M+ image–text pairs) for multimodal fusion research.

Hardware, software, and implementation. All experiments are conducted on an NVIDIA A100 GPU (40 GB memory) with 48 CPU cores and 512 GB RAM. Image

TABLE I: NutriDermaI Pipeline Stage Overview

Stage	Input / Output	Dataset (Train/Val/Test)	Preprocessing	Model	Status	Key Metric
1	Raw image / text	N/A	Format validation, metadata extraction	Centralised normaliser	In Prog.	—
2 (SLS)	Image → Mask	ISIC 2016–17 (2.9K/400/979)	Resize 512×512, norm., hair removal	ARC-UNet	✓	Dice: 0.8506 Bdry F1: 0.8815
3 (SLRC)	Mask + Image → ROI	ISIC 2016–17 (2.9K/400/979)	Morphology, connected components, bbox, 10% pad	OpenCV 4.5	✓	Coverage: 98.4%
4 (HC)	ROI → Label + Features	ISIC + HAM + ISIC 19–24 (35K/5K/10K)	Augmentation, normalisation, MixUp	EfficientFormer-V2-S2	✓	Acc: 94.7% (bin.) Acc: 89.2% (7-cl.)
5 (MedProc)	Report → Symptoms + E	MIMIC-III/IV (50K/5K/10K)	OCR, cleanup, tokenisation (≤ 512 tokens, stride 100)	BioBERT	✓	NER F1: 0.87–0.89
6 (Fusion)	HC + MedProc → Clinical JSON	Derm1M + MIMIC (700K/100K/200K)	Feature alignment, z-norm, embedding	MARIA Transformer	In Prog.	Risk AUC (TBD)
7 (VQA)	JSON + ROI + Q → Answer	DermaVQA + PathVQA (80K/10K/15K)	Prompt conversion, tokenisation	R-LLaVA	In Prog.	VQA F1 (TBD)

stages (SLS, SLRC, HC) are implemented in PyTorch 2.1 with OpenCV for image processing. Clinical NLP (MedProc) uses HuggingFace Transformers with BioBERT-base. All code is containerised using Docker for reproducibility.

Hyperparameter configurations and training details. Key training configurations are detailed below:

- **SLS (ARC-UNet):** Batch size 8, learning rate 10^{-4} , optimizer Adam, loss hybrid Dice+CE (ratio 0.6/0.4), 100 epochs, early stopping patience 10, L_2 regularisation $\lambda = 10^{-5}$. Augmentation: random flips, rotations $\pm 15^\circ$, elastic deformations.
- **HC (EfficientFormerV2-S2):** Batch size 16×2 (gradient accumulation), learning rate 10^{-3} (head) / 10^{-4} (backbone), OneCycleLR scheduler, loss AsymmetricFocalLoss ($\gamma = 2.0$), 80 epochs, weighted sampling for class balance, MixUp/CutMix augmentation.
- **MedProc (BioBERT):** Batch size 32, learning rate 5×10^{-5} , optimizer AdamW, 10 epochs, 512-token limit with stride-100 chunking, dropout 0.1, label smoothing 0.1. Evaluation uses ECE (Expected Calibration Error) and per-entity F1 scores.

We present stage-wise results for the four completed stages (2–5). For each stage, we document the dataset, algorithm, training configuration, and measured performance metrics.

B. Stage 2: Skin Lesion Segmentation — ARC-UNet

Dataset. ISIC 2016–2017: 4,279 images (2,900 train / 400 val / 979 test).

Algorithm. ARC-UNet with squeeze-and-excitation attention, residual connections, and an asymmetric encoder–decoder.

TABLE II: Stage 2 (SLS) — ARC-UNet Segmentation Results

Metric	Single-Pass	TTA (7-view)	Gain
Dice Coefficient	0.8506	0.8512	+0.07%
IoU	0.7718	0.7726	+0.10%
Pixel Accuracy (%)	93.70	93.75	+0.05%
Boundary F1-Score	0.8801	0.8815	+0.16%
Sensitivity (Recall)	0.9102	0.9118	+0.18%
Specificity	0.9756	0.9761	+0.05%
<i>Optimal threshold: 0.44 (well-calibrated)</i>			

Training. Hybrid Dice+CE loss (ratio 0.6/0.4), Adam (LR = 10^{-4}), batch size 8, 100 epochs, L_2 regularisation ($\lambda = 10^{-5}$), early stopping (patience 10), augmentation (flips, rotations $\pm 15^\circ$, elastic deformations).

Results. Table II reports single-pass and 7-view test-time augmentation (TTA) performance. ARC-UNet achieves Boundary F1 of 0.8815 with balanced sensitivity and specificity. Single-pass inference is recommended for clinical deployment.

C. Stage 3: Lesion ROI Cropping — OpenCV Pipeline

Dataset. Derived from ISIC 2016/2017 SLS output masks (4,279 images; no separate dataset).

Algorithm. Deterministic four-step pipeline:

- 1) *Morphological cleanup:* binary closing (disk, $r = 3$) to fill segmentation holes.
- 2) *Connected-component labelling:* 8-connectivity; retain the largest component.
- 3) *Bounding box extraction:* axis-aligned bounding box with 10% padding on all sides.

TABLE III: Stage 3 (SLRC) — ROI Cropping Results

Metric	Value
Mean Lesion Coverage (%) pixels retained)	98.4% $\pm$ 1.2%
Border Preservation (%) contour within ROI)	96.7% $\pm$ 2.1%
Mean ROI Size Variation (vs. anat. diam.)	0.34 mm (IQR: 0.18–0.65 mm)
Computational Time per ROI (CPU)	12.3 ms

- 4) *Aspect-ratio-preserving resize*: bilinear interpolation to 224×224 with black-border padding if needed.

Results. Table III confirms high lesion coverage (98.4%) and border preservation (96.7%) at 12.3 ms latency per ROI.

D. Stage 4: Hierarchical Classification — EfficientFormerV2-S2

Dataset. ISIC 2018 + HAM10000 + ISIC 2019–2024; 50,000 balanced samples (70/10/20 split, no patient overlap).

Algorithm. EfficientFormerV2-S2 with dual-head classification (benign/malignant + 7-class) and parallel ABCD sub-pipelines.

Training. AsymmetricFocalLoss ($\gamma = 2.0$), OneCycleLR (head 10^{-3} , backbone 10^{-4}), batch 16×2 gradient accumulation, 80 epochs, MixUp/CutMix, WeightedRandomSampler.

Results. Table IV shows that explicit ABCD features improve both interpretability and performance. TTA provides approximately 1% improvement for critical cases; single-pass is acceptable for routine triage.

TABLE IV: Stage 4 (HC) — Hierarchical Classification Results

Metric	Single-Pass	TTA (7-view)	Gain
<i>Binary Classification (Benign / Malignant)</i>			
Accuracy (%)	94.7	95.3	+0.6%
Sensitivity	0.9512	0.9618	+1.06%
Specificity	0.9387	0.9425	+0.38%
AUC-ROC	0.9751	0.9803	+0.52%
<i>Multi-Class Classification (7 Categories)</i>			
Accuracy (%)	89.2	90.1	+0.9%
Weighted F1-Score	0.8876	0.8979	+1.03%
Macro F1-Score	0.8654	0.8762	+1.08%
Per-Class Balanced Acc.	0.8905	0.8998	+0.93%

E. Stage 5: Medical Information Processing — BioBERT

Dataset. MIMIC-III/IV: 50K train / 5K val / 10K test (no patient overlap).

Algorithm. BioBERT-base augmented with two transformer blocks for clinical NER, relation extraction, and rule-based Evolution (E) scoring.

Training. AdamW (LR = 5×10^{-5}), batch 32, 10 epochs, 512-token limit with stride-100 chunking, dropout 0.1, label smoothing 0.1.

Results. Table V shows NER F1 of 0.87–0.89 across clinical entity types, with well-calibrated uncertainty (ECE 0.0201), suitable for downstream fusion.

TABLE V: Stage 5 (MedProc) — Clinical NLP Results

Task / Entity	Prec.	Rec.	F1
<i>Named Entity Recognition (NER)</i>			
Symptoms	0.8901	0.8596	0.8742
Body sites	0.9087	0.8796	0.8934
Temporal mentions	0.8312	0.8011	0.8156
Severity indicators	0.7956	0.7698	0.7821
<i>Relation Extraction (Symptom-to-Evolution)</i>			
Progression detected (P / R)	0.8764	0.8523	—
AUC-ROC		0.8934	
<i>Evolution (E) Score Quality</i>			
Expected Calibration Error (ECE)		0.0201	
Mean Absolute Error (scale 0–1)		0.1547	

F. Integration and Data Flow Validation

End-to-end integration tests across Stages 2–5 confirm correct data flow: SLS mask (512×512 binary) → SLRC ROI (224×224 RGB) → HC classification logits and ABCD features + MedProc JSON (symptoms and E-score). Structured outputs—JSON for MedProc, numpy arrays for image stages—are properly serialised and deserialised between stages. Per-sample latency on an NVIDIA A100 is: SLS 280 ms, SLRC 12 ms, HC 45 ms, MedProc 180 ms; total **517 ms**, which is acceptable for clinical batch processing.

VII. DISCUSSION

A. Validation of Modular Design and Component Performance

The four-stage modular architecture yields several practical advantages. First, each stage can be evaluated independently on established benchmarks (ISIC for SLS/HC; MIMIC for MedProc; deterministic validation for SLRC), facilitating transparent reporting and reproducibility. SLS achieves Dice 0.8506 and Boundary F1 0.8815, competitive with recent state-of-the-art U-Net variants. HC attains 94.7% binary accuracy and 89.2% 7-class accuracy with strong per-class balance, demonstrating that explicit ABCD feature extraction does not compromise classification performance. MedProc achieves NER F1 of 0.87–0.89 with low calibration error (ECE 0.0201), suitable for downstream fusion. Together, these results confirm that modular independence does not compromise individual component quality.

Second, SLS and HC are deployable as standalone components, with MedProc available as an optional enrichment layer. This flexibility supports iterative deployment: institutions may begin with image-only analysis (SLS + HC) and incrementally integrate text processing as clinical workflows mature. Third,

all outputs are structured (JSON or numpy arrays) with calibrated confidence scores, enabling explainability and audit trails. Fourth, the microservice-compatible design supports containerised cloud and on-premise deployment with role-based access control.

B. Implications of ABCDE Computation in IT-Fusion

A key architectural choice is to compute ABCD scores in Stage 6 (IT-Fusion) rather than Stage 4 (HC). This design decision reflects two insights: (i) ABCD metrics cannot be properly calibrated using image features alone, as clinician-annotated ABCDE labels typically reflect integrated judgement combining visual, temporal, and contextual factors; and (ii) Evolution (E) depends exclusively on clinical text and cannot be inferred from images. By deferring ABCD score computation to the fusion stage, NutriDermAI ensures that all five criteria are computed consistently and that the final risk estimate properly integrates both modalities. Future work may explore learning-based ABCD calibration mappings via annotation of a prospective cohort.

C. Safety and Ethical Considerations

NutriDermAI is designed as a decision support tool, not a standalone diagnostic system. All outputs include explicit confidence levels, and low-confidence predictions automatically trigger escalation pathways to human dermatologists. The structured Clinical JSON audit trail documents which image and text features contributed to each decision, enabling post-hoc review and quality assurance.

Deployment must comply with emerging regulatory frameworks for AI-based medical devices. The FDA's recent guidance on AI/ML-based Software as a Medical Device (SaMD) [61] emphasises transparency, performance monitoring, and post-market surveillance. The EU AI Act classifies medical AI as high-risk, requiring documented risk assessments, data governance, and human oversight [62]. NutriDermAI's modular design and structured outputs facilitate compliance: each stage can be independently validated, and the JSON evidence trail enables detailed audits.

Fairness and bias are critical concerns. The current training cohorts (ISIC, HAM10000) are known to skew towards lighter skin tones, potentially reducing performance on darker skin phenotypes [?], [?]. Prospective fairness audits across skin tones, anatomical sites, and lesion types are planned. If disparities are detected, targeted data augmentation, fine-tuning on diverse cohorts, and ensemble methods can mitigate bias. Multinational clinical validation with diverse patient populations is essential before broad deployment.

D. Key Strengths and Architectural Contributions

The NutriDermAI architecture demonstrates several distinct strengths that differentiate it from existing dermatology AI systems:

- **Multimodal integration.** By combining dermoscopic images, clinical text, and user queries, the system captures information that no single modality can provide alone.

This fusion enables more robust and contextually grounded clinical reasoning.

- **Structured explainability.** The Clinical Knowledge JSON serves as an intermediate representation that makes all reasoning transparent. Clinicians can audit which image and text features contributed to each ABCDE score and risk decision, supporting trust and compliance.
- **ABCDE-aware reasoning.** Unlike generic medical VQA systems, our approach explicitly computes and reasons over the ABCDE rule—the clinical gold standard for melanoma risk assessment. This alignment with dermatology practice standards improves clinical relevance and uptake.
- **Knowledge-grounded conversational AI.** By integrating Retrieval-Augmented Generation (RAG), the system grounds responses in verified medical guidelines and dermatology evidence, reducing hallucinations and improving accuracy compared to LLM-only approaches.
- **Modular design for iterative deployment.** Each stage is independently validated and can be improved or replaced independently. Institutions can start with image-only analysis (SLS + HC) and progressively integrate text processing and conversational reasoning as clinical workflows mature.
- **Calibrated confidence and clinical escalation.** Every stage reports uncertainty estimates; low-confidence predictions automatically trigger escalation pathways to human dermatologists, ensuring patient safety.

Four core modules are fully implemented and evaluated:

- **Stage 2 (SLS):** Dice 0.8506, IoU 0.7718, 93.70% pixel accuracy (Table II).
- **Stage 3 (SLRC):** 98.4% coverage, 96.7% border preservation, 12.3 ms latency (Table III).
- **Stage 4 (HC):** 94.7% binary accuracy (TTA 95.3%), 89.2% 7-class accuracy (TTA 90.1%) (Table IV).
- **Stage 5 (MedProc):** NER F1 0.8742–0.8934, ECE 0.0201 (Table V).

Remaining stages requiring implementation are: Stage 1 (input normalisation), Stage 6 (IT-Fusion multimodal transformer with ABCDE calibration), and Stage 7 (R-LLaVA VQA). Stages 8–9 (conversational LLM with RAG, and deployment) are future work pending completion of Stages 1 and 6–7. Integration and validation of the complete pipeline is targeted for Q4 2026.

E. Generalisability and Deployment Considerations

The pipeline's modular design supports generalisation across settings. ISIC benchmarks span multiple imaging devices and geographies, suggesting reasonable device robustness. However, real-world deployment in clinical settings will likely encounter out-of-distribution inputs (poor lighting, large artefacts, non-standard cameras). Robust out-of-distribution detection and failure mode analysis are planned. Furthermore, sequential processing (SLS → SLRC → HC → MedProc → Fusion) introduces latency. Optimisations include batch processing, model quantisation, and edge deployment of lightweight models for resource-limited settings.

The English-only text processing is a significant limitation for international deployment. Multilingual clinical NLP models and translated prompts will extend applicability. Stateless inference (processing each case independently) also limits clinical utility; a stateful version maintaining longitudinal ABCDE trends within a patient’s record is planned for integration with clinical EHR systems.

VIII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

The NutriDermAI prototype, though comprehensive in design, carries several known limitations that inform the research roadmap:

- **English-only text processing.** All clinical NLP components are trained on English-language corpora. In global health contexts where patients may provide reports in other languages, a multilingual extension is necessary [63].
- **Skin tone bias.** Like many dermatology datasets, the training cohorts (ISIC, HAM10000) skew towards lighter skin tones. Systematic fairness audits across diverse skin tones are planned [?], [?].
- **Dataset curation gaps.** The current HC stage uses curated ISIC + HAM10000 data; in-the-wild deployment may encounter artefacts (hair, ink, lighting) not well-represented in training sets. Robust out-of-distribution detection is planned.
- **Stateless inference.** The current pipeline processes each image independently without longitudinal context. Clinical deployment requires temporal patient records; a stateful version maintaining historical ABCDE trends is planned.
- **Staged validation.** While Stages 2–5 have been independently validated, end-to-end performance of the fused system (Stages 6–7) is pending completion and clinical evaluation.

B. Future Work and Clinical Validation Roadmap

- **Completion of core pipeline stages.** Stage 1 (input normalisation), Stage 6 (IT-Fusion multimodal transformer), and Stage 7 (R-LLaVA VQA) are in active development. Integration and validation are expected by Q4 2026.
- **Prospective clinical validation.** Once Stages 1–7 are complete, a prospective study comparing NutriDermAI’s ABCDE scores and risk rankings against board-certified dermatologist consensus will establish diagnostic concordance and clinical utility [?].
- **Fairness and bias audits.** Systematic evaluation across skin tones, lesion types, and anatomical sites will be conducted using established fairness metrics [?], [?].
- **Regulatory compliance and deployment.** Compliance with FDA guidance on AI/ML-based Software as a Medical Device (SaMD) [61] and the EU AI Act [62] will be undertaken. Stage 9 deployment will include role-based access, audit logging, data encryption, and continuous monitoring for model drift and failure modes.
- **Conversational robustness and safety.** Stage 8 (conversational LLM with RAG) will be stress-tested for hallucination, adversarial prompts, and out-of-scope questions.

Guardrails following AMIE and clinical chatbot best practices [42], [45] will be implemented.

- **Multi-language and cross-cultural adaptation.** Multilingual clinical NLP models and culturally sensitive VQA prompts will extend NutriDermAI’s applicability to low-resource regions.
- **Real-time inference optimisation.** Model distillation and quantisation will enable deployment on resource-constrained devices (smartphones, rural clinics with limited bandwidth).

IX. CONCLUSION

This paper presents NutriDermAI, a modular, ABCDE-aware multimodal dermatology decision support system with four validated core stages: skin lesion segmentation (Dice 0.8506, Boundary F1 0.8815), ROI extraction (98.4% coverage, 12.3 ms), hierarchical classification (94.7% binary accuracy, 89.2% 7-class), and clinical NLP for evolution scoring (NER F1 0.87–0.89, ECE 0.0201). End-to-end integration across these stages is confirmed with a total pipeline latency of 517 ms on an NVIDIA A100.

Core architectural contribution. The critical contribution of NutriDermAI is the principled fusion of image and text modalities to compute clinically calibrated ABCDE scores, a task typically performed ad-hoc by dermatologists with significant inter-observer variation. By routing all downstream reasoning through a structured Clinical Knowledge JSON, the system provides interpretability, independent stage evaluation, and redeployable components suitable for clinical governance and audit. Furthermore, by integrating knowledge-grounded conversational AI via Retrieval-Augmented Generation, NutriDermAI supports multi-turn clinical dialogue while minimising hallucination and ensuring alignment with verified medical evidence.

Thesis statement. The proposed architecture bridges perception, reasoning, and interaction by integrating multimodal learning, structured clinical inference, and knowledge-grounded conversational AI, thereby enabling a scalable and explainable dermatology decision support system. This integrated design addresses a critical gap in the literature, where existing systems either provide image-only classification without clinical context, or offer generic medical VQA without ABCDE-aware reasoning. By combining all three capabilities in a unified, modular pipeline, NutriDermAI positions itself as a comprehensive platform for both clinical decision support and patient education.

Path forward. NutriDermAI provides a validated research foundation for ABCDE-aware dermatology AI systems that combine strong empirical performance with transparency and modular reusability. Prospective clinical validation studies, completion of Stages 1, 6, and 7, and fairness audits across diverse populations are the immediate priorities. If such validation confirms diagnostic concordance with board-certified dermatologists, NutriDermAI could serve as a scalable triage tool in resource-limited settings where specialist access is restricted [47], while simultaneously supporting clinical training and reducing diagnostic variability in high-income regions.

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